



# React JS Full Stack Development with PostgreSQL

*(React JS + Express JS + PostgreSQL + JWT + Redux Toolkit + Razorpay)*



## Course Duration

**1.5 Hrs to 2 Hrs / Session**

**5 Days / Week**

**30-35 Days / 45-50 Hours (Flexible)**



## Course Objective

This industry-oriented hands-on Training Program is designed to help students build modern, production-ready web applications using React JS and the PERN Stack. The course starts with JavaScript ES6 fundamentals and gradually progresses to authentication, REST APIs, state management, payment integration, and a complete E-Commerce application.

By the end of this course, participants will be able to design, develop, and deploy full-stack React applications using industry best practices.

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## Course Modules

### Module 1 – JavaScript ES6+ Fundamentals

Build a strong foundation in modern JavaScript.

#### Topics Covered

- Variables (`let` & `const`)
- Arrow Functions
- Template Literals
- Destructuring
- Spread & Rest Operators
- ES Modules
- Promises

- Async/Await

#### **Outcome**

- ✓ Write clean, modern JavaScript required for React development.
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## **Module 2 – React JS Core Concepts**

Understand how React builds modern Single Page Applications.

#### **Topics Covered**

- Introduction to React
- SPA Architecture
- JSX
- Components
- Functional Components
- Props
- State
- Event Handling
- Conditional Rendering
- Lists & Keys

#### **Outcome**

- ✓ Build reusable and interactive React components.
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## **Module 3 – React Hooks**

Learn React's most powerful feature.

#### **Topics Covered**

- useState
- useEffect
- useRef
- useMemo
- useCallback

- useContext
- useReducer
- useEffect
- Custom Hooks
- useFetch
- useHttp
- useToggle
- useLocalStorage

### **Outcome**

- ✓ Manage state, side effects, performance optimization, and reusable logic.
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## **Module 4 – React Router**

Create professional multi-page applications.

### **Topics Covered**

- BrowserRouter
- Route Configuration
- Navigation using Link
- Dynamic Routes
- Route Parameters
- Programmatic Navigation

### **Outcome**

- ✓ Build complete navigation systems for React applications.
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## **Module 5 – Authentication & Authorization**

Implement secure user authentication.

### **Topics Covered**

- Login
- Registration

- JWT Authentication
- Protected Routes
- Role-Based Authorization
- Admin & User Access Control

#### **Outcome**

- ✓ Secure React applications using authentication and authorization.
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## **Module 6 – Backend Development with Express JS & PostgreSQL**

Build REST APIs for React applications.

#### **Topics Covered**

- Express Server Setup
- PostgreSQL Database
- REST API Development
- CRUD Operations
- Database Integration
- Axios Communication

#### **Outcome**

- ✓ Connect React applications with a backend API and PostgreSQL database.
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## **Module 7 – Real-World JWT Authentication**

Implement production-level authentication flow.

#### **Topics Covered**

- JWT Access Tokens
- Refresh Tokens
- Token Expiration
- Axios Interceptors

- Automatic Token Refresh
- Session Management
- Protected APIs

#### **Outcome**

- ✓ Understand and implement secure authentication used in enterprise applications.
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## **Module 8 – State Management using Redux Toolkit**

Manage application-wide state efficiently.

#### **Topics Covered**

- Redux Store
- Slices
- Reducers
- createAsyncThunk
- Async API Calls
- Global State Management

#### **Outcome**

- ✓ Build scalable React applications using Redux Toolkit.
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## **Module 9 – API Integration & UI Development**

Build professional-looking applications.

#### **Topics Covered**

- Axios Configuration
- Environment Variables
- API Integration

- Tailwind CSS
- Responsive Design
- Loading Indicators
- Toast Notifications
- SweetAlert2
- Error Handling

### Outcome

- ✓ Create responsive and user-friendly React applications.
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## Module 10 – Capstone Project



### E-Commerce Application

Develop a complete real-world project from scratch.

### Features

- User Registration
- Login System
- JWT Authentication
- Product Management
- Shopping Cart
- Razorpay Payment Integration
- Orders Management
- Admin Panel
- Protected APIs
- Responsive UI
- PostgreSQL Database
- Express REST APIs

### Outcome

- ✓ Gain practical experience by building a complete production-ready Full Stack application.
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### Technologies Covered

## ■ Additional Industry-Oriented Topics

The following modules extend the core React JS + Express JS + PostgreSQL training with practical, industry-oriented skills.

### Module 11 – AI/LLM Integration

Learn how to add simple AI features to your full-stack applications using an LLM API.

#### Topics Covered

- Introduction to AI & Large Language Models (LLMs)
- LLM APIs and API Integration
- Sending Prompts and Receiving Responses
- Building a simple AI-powered feature in a full-stack application
- Calling an LLM API securely from the backend

**Outcome: ✓ Add practical AI capabilities to full-stack applications.**

### Module 12 – Advanced PostgreSQL

Learn how to handle real-world data with joins, relationships, indexes, and pagination.

#### Topics Covered

- Table Relationships
- INNER JOIN, LEFT JOIN and related queries
- Primary Keys and Foreign Keys
- Indexes and query performance
- Pagination and efficient data retrieval
- Working with real-world relational data

**Outcome: ✓ Work efficiently with relational data and build scalable database queries.**

### Module 13 – Backend Architecture & Security

Learn how to structure and secure your backend like a real-world application.

#### Topics Covered

- Layered / Modular Backend Architecture
- Controllers, Services and Data Access
- Environment Variables and Configuration
- Input Validation
- Authentication and Authorization
- API Security Best Practices
- Secure handling of sensitive information

**Outcome: ✓ Structure and secure backend applications using real-world practices.**

## Module 14 – Testing & Debugging

Learn how to test your code and find and fix common problems.

### Topics Covered

- Introduction to Testing
- Testing frontend and backend functionality
- API Testing
- Debugging React applications
- Debugging Express.js APIs
- Browser DevTools and Network debugging
- Identifying and fixing common application errors

**Outcome: ✓ Test, debug and troubleshoot full-stack applications systematically.**

## Module 15 – Git, GitHub & Deployment

Learn how to manage your code with Git and put your project live.

### Topics Covered

- Git fundamentals
- Repositories, commits and branches
- Merging and resolving conflicts
- GitHub repositories and collaboration
- Pushing and managing project code
- Production build and deployment concepts
- Deploying a full-stack application

**Outcome: ✓ Manage source code professionally and deploy a full-stack project.**

- HTML5
  - CSS3
  - JavaScript (ES6+)
  - React JS
  - React Router
  - React Hooks
  - Redux Toolkit
  - Axios
  - Tailwind CSS
  - Express JS
  - Node.js
  - PostgreSQL
  - JWT Authentication
  - Cookie-Based Authentication
  - REST APIs
  - Razorpay Payment Gateway
  - Git & GitHub (Optional)
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## Learning Outcomes

After completing this course, participants will be able to:

- Develop modern React applications from scratch.
  - Build reusable and scalable React components.
  - Consume REST APIs using Axios.
  - Develop backend services with Express.js.
  - Integrate PostgreSQL databases.
  - Implement secure JWT authentication.
  - Manage global application state using Redux Toolkit.
  - Build responsive UIs with Tailwind CSS.
  - Integrate online payment gateways.
  - Develop and deploy a complete Full Stack E-Commerce application.
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## Who Should Attend?

- Students
- Fresh Graduates
- Software Engineers

- Frontend Developers
  - Full Stack Aspirants
  - Professionals transitioning to React
  - Anyone interested in modern web development
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## Prerequisites

- Basic Computer Knowledge
  - Basic HTML & CSS
  - Basic JavaScript Knowledge (preferred but not mandatory)
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## Course Deliverables

- ✓ Complete Source Code
- ✓ Hands-on Coding Sessions
- ✓ Real-Time Project Development
- ✓ Interview-Oriented Explanations
- ✓ Industry Best Practices
- ✓ Course Completion Certificate (*Optional*)
- ✓ Lifetime Project Reference